

Finite-facet bodies and the centered-simplex test case for a Minkowski-sum gauge question

Candidate partial result for arXiv:1811.12548

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Status. Candidate partial result, likely valid. This packet proves the conjectured order for centered polytopes with linearly many facets and determines the exact first-order asymptotic for centered simplices. It does not settle the universal worst-case question over all centered convex bodies.

1 Source question

Huang, Slomka, Tkocz, and Vritsiou ask in Section 6 of [1] whether, for independent uniform random vectors X, Y on a centered convex body $K \subset \mathbb{R}^n$, there is a universal $\alpha < 1$ for which

$$\mathbb{E}\|X + Y\|_K \leq 2 - \Omega(n^{-\alpha}),$$

and especially whether the worst deficit is of order $n^{-1/2}$. Here $\|z\|_K = \inf\{t > 0 : z \in tK\}$ is the Minkowski functional; it need not be a norm when K is nonsymmetric. The source computes for the cube Q_n that

$$\mathbb{E}\|X + Y\|_{Q_n} = 2 - \frac{2 \cdot 4^n}{(2n+1) \binom{2n}{n}} = 2 - \sqrt{\frac{\pi}{n}} + o(n^{-1/2}). \quad (1)$$

Figure 1 reproduces the complete source question from page 15.

2 Result

Theorem 1 (Finite-facet bound). *Let $K \subset \mathbb{R}^n$ be a polytope whose barycenter is the origin and which has N facets. If X, Y are independent and uniform on K , then*

$$\mathbb{E}\|X + Y\|_K \leq 2 - \frac{3}{16\sqrt{N}}. \quad (2)$$

Consequently, for every fixed $A > 0$, the source's proposed $n^{-1/2}$ deficit holds uniformly over centered polytopes with at most An facets.

Theorem 2 (Centered-simplex asymptotic). *Let $K_n \subset \mathbb{R}^n$ be any simplex whose barycenter is the origin, and let X, Y be independent and uniform on K_n . Then*

$$\mathbb{E}\|X + Y\|_{K_n} = 2 - \sqrt{\frac{\pi}{2n}} + o(n^{-1/2}). \quad (3)$$

Consequently, the ratio of the simplex deficit to the cube deficit tends to $1/\sqrt{2}$. In particular, for all sufficiently large n ,

$$\mathbb{E}\|X + Y\|_{K_n} > \mathbb{E}\|X + Y\|_{Q_n}.$$

$$\frac{\mathbb{P}\left(\frac{X+Y}{2} \in rK\right)}{\mathbb{P}(X \in rK)} \geq (1+c)^n,$$

where X and Y are independent random vectors, uniformly distributed on K .

We remark that the above conjecture implies an exponentially better upper bound for Hadwiger’s covering problem. Moreover, the conjecture seems interesting in its own right and attempts in a way to quantify the intuition that the convolution of a uniform distribution with itself looks already more like a “bell curve” than like the flat distribution it originates from.

Another question that would capture this if answered in the affirmative is the following. Let X and Y be independent random vectors, uniformly distributed on a centred convex body K . Is it true that $\mathbb{E}\|X+Y\|_K \leq 2 - \Omega(n^{-\alpha})$ with $\alpha \in [0, 1)$ independent of K ? Or rather, given such an α , for which convex bodies in $\mathbb{R}^n$ does this bound hold? (Note that any such bound would improve on the trivial upper bound coming from the triangle inequality, which totally neglects independence: $\mathbb{E}\|X+Y\|_K \leq 2\mathbb{E}\|X\|_K = 2(1 - \frac{1}{n+1})$.) In the previous section we proved that $\mathbb{E}\|X+Y\|_K$ is indeed upper-bounded by a constant smaller than 2 for convex bodies with a positive modulus of convexity. For the cube however, it can be checked that $\mathbb{E}\|X+Y\|_K = 2(1 - \frac{4^n}{(2n+1)\binom{2n}{n}}) \sim 2 - \sqrt{\frac{\pi}{n}}$. Thus, we can also ask whether, in general, the bound $2 - \Omega(n^{-1/2})$ is the worst case. If this is true, it would give another proof for our Main Theorems [1.1](#) and [1.3](#).

Figure 1: The question and cube benchmark in Section 6 of [\[1\]](#).

Proof intuition

For a polytope the gauge is the maximum of its normalized facet functionals. The deficit from a fixed facet is a nonnegative log-concave random variable of mean one, and such a variable has probability at most $2t$ of lying in $[0, t]$. For the sum of two independent points to have gauge within t of two, both points must lie within height t of a common facet. The resulting probability is at most $4Nt^2$, yielding Theorem [1](#) at $t \asymp N^{-1/2}$.

Affine invariance lets us put the simplex in barycentric coordinates. A uniform point is then a flat Dirichlet vector shifted so that the barycenter is zero, and the simplex gauge is simply minus the smallest coordinate. The problem therefore becomes an extreme-value calculation for $\min_i(P_i + Q_i)$.

Flat Dirichlet coordinates are normalized independent exponentials. Their normalizing sums concentrate, so the relevant minimum is asymptotically the minimum of $n+1$ independent Gamma(2, 1) variables. The Gamma(2, 1) distribution has quadratic lower tail $z^2/2$; therefore its minimum has Rayleigh scaling and limiting mean $\sqrt{\pi/2}$. A uniform tail bound and an exponentially small normalization bad event justify convergence of expectations. The two theorems still supply no universal comparison for bodies with arbitrary or infinitely many supporting facets.

3 Proof

3.1 The finite-facet theorem

We first isolate the one-dimensional estimate.

Lemma 1 (Log-concave boundary small ball). *Let U be a nonnegative random variable with a log-concave density and $\mathbb{E}U = 1$. Then, for $0 \leq t \leq 1/2$,*

$$\mathbb{P}(U \leq t) \leq 2t.$$

Proof. Let $S(t) = \mathbb{P}(U \geq t)$. The survival function of a log-concave density is log-concave, so $H = -\log S$ is convex and $H(0) = 0$. For $u \geq t > 0$, convexity gives $H(u) \geq (u/t)H(t)$. If $H(t) = 0$ the conclusion is immediate, so assume $H(t) > 0$. Then

$$1 = \mathbb{E}U = \int_0^\infty S(u) du \leq t + \int_t^\infty e^{-(u/t)H(t)} du \leq t + \frac{t}{H(t)}.$$

Thus $H(t) \leq t/(1-t) \leq 2t$, and consequently

$$\mathbb{P}(U \leq t) = 1 - e^{-H(t)} \leq H(t) \leq 2t.$$

The endpoint $t = 0$ follows by continuity. □

Proof of Theorem 1. Choose normalized facet functionals $\varphi_1, \dots, \varphi_N$ such that

$$K = \{x : \varphi_j(x) \leq 1 \text{ for } 1 \leq j \leq N\}.$$

Since the origin is in the interior, $\|z\|_K = \max_j \varphi_j(z)$. Put

$$U_j = 1 - \varphi_j(X), \quad V_j = 1 - \varphi_j(Y).$$

Each variable is nonnegative. The one-dimensional marginal of uniform measure on a convex body is log-concave, and the barycenter assumption gives $\mathbb{E}\varphi_j(X) = 0$; hence $\mathbb{E}U_j = 1$. Lemma 1 applies.

The following deficit identity is exact:

$$2 - \|X + Y\|_K = \min_{1 \leq j \leq N} (U_j + V_j) =: D. \tag{4}$$

For $0 \leq t \leq 1/2$, independence and a union bound give

$$\begin{aligned} \mathbb{P}(D \leq t) &\leq \sum_{j=1}^N \mathbb{P}(U_j + V_j \leq t) \\ &\leq \sum_{j=1}^N \mathbb{P}(U_j \leq t) \mathbb{P}(V_j \leq t) \leq 4Nt^2. \end{aligned}$$

Take $t_0 = 1/(4\sqrt{N})$. Then $\mathbb{P}(D > t_0) \geq 3/4$, so

$$\mathbb{E}D \geq t_0 \mathbb{P}(D > t_0) \geq \frac{3}{16\sqrt{N}}.$$

Together with (4), this proves (2). □

3.2 The exact simplex asymptotic

Put $m = n + 1$ and work in the hyperplane

$$H_m = \{x \in \mathbb{R}^m : \sum_{i=1}^m x_i = 0\}.$$

Every centered n -simplex is a linear image of

$$K = \{x \in H_m : x_i \geq -1 \text{ for every } i\}.$$

Both uniform measure and the Minkowski functional commute with invertible linear maps, so it suffices to treat this K .

Let $P = (P_1, \dots, P_m)$ and $Q = (Q_1, \dots, Q_m)$ be independent and uniform on the probability simplex

$$\Delta_{m-1} = \{p_i \geq 0 : \sum_i p_i = 1\}.$$

Then $X = mP - \mathbf{1}$ and $Y = mQ - \mathbf{1}$ are independent and uniform on K . For $z \in H_m$,

$$\|z\|_K = \inf\{t > 0 : z_i \geq -t \ \forall i\} = -\min_i z_i.$$

It follows exactly that

$$\|X + Y\|_K = 2 - mM_m, \quad M_m := \min_{1 \leq i \leq m} (P_i + Q_i). \quad (5)$$

We use the exponential representation. Let E_i, F_i be mutually independent exponential random variables of mean one, and set

$$S_m = \sum_{i=1}^m E_i, \quad T_m = \sum_{i=1}^m F_i.$$

Then

$$(P_i)_{i=1}^m \stackrel{d}{=} (E_i/S_m)_{i=1}^m, \quad (Q_i)_{i=1}^m \stackrel{d}{=} (F_i/T_m)_{i=1}^m.$$

First ignore the two random denominators. The variables $Z_i := E_i + F_i$ are independent Gamma(2, 1) variables, with survival function

$$q(z) := \mathbb{P}(Z_i > z) = e^{-z}(1 + z).$$

Define

$$W_m := \sqrt{m} \min_{1 \leq i \leq m} Z_i.$$

For every fixed $t \geq 0$,

$$\begin{aligned} \mathbb{P}(W_m > t) &= q(t/\sqrt{m})^m \\ &= \left(1 - \frac{t^2}{2m} + O(m^{-3/2})\right)^m \longrightarrow e^{-t^2/2}. \end{aligned}$$

Thus W_m converges in distribution to the Rayleigh random variable W with survival function $\mathbb{P}(W > t) = e^{-t^2/2}$.

We record the uniform tail control needed below. Since

$$z - \log(1 + z) \geq z^2/4 \quad (0 \leq z \leq 1)$$

and

$$z - \log(1+z) \geq (1 - \log 2)z \quad (z \geq 1),$$

we have, with $c_0 = 1 - \log 2 > 0$,

$$\mathbb{P}(W_m > t) \leq \begin{cases} e^{-t^2/4}, & 0 \leq t \leq \sqrt{m}, \\ e^{-c_0 t \sqrt{m}}, & t \geq \sqrt{m}. \end{cases} \quad (6)$$

These bounds give a common integrable majorant for the survival functions and also uniformly bound $\mathbb{E}W_m^2$. Hence, by the survival-integral formula and dominated convergence,

$$\mathbb{E}W_m \longrightarrow \mathbb{E}W = \int_0^\infty e^{-t^2/2} dt = \sqrt{\frac{\pi}{2}}. \quad (7)$$

It remains to restore the denominators. Put $\delta_m = m^{-1/4}$ and let

$$G_m = \{|S_m/m - 1| \leq \delta_m, |T_m/m - 1| \leq \delta_m\}.$$

Standard exponential Chernoff bounds, obtained directly from $\mathbb{E}e^{\theta E_i} = (1 - \theta)^{-1}$, give

$$\mathbb{P}(G_m^c) \leq 4e^{-c\sqrt{m}} \quad (8)$$

for a universal $c > 0$ and all sufficiently large m . On G_m , coordinate by coordinate,

$$\frac{E_i + F_i}{m(1 + \delta_m)} \leq \frac{E_i}{S_m} + \frac{F_i}{T_m} \leq \frac{E_i + F_i}{m(1 - \delta_m)}.$$

Consequently, if

$$A_m := m^{3/2}M_m,$$

then on G_m

$$\frac{W_m}{1 + \delta_m} \leq A_m \leq \frac{W_m}{1 - \delta_m}. \quad (9)$$

Finally, $M_m \leq 2/m$ because the minimum of the m numbers $P_i + Q_i$ is no larger than their average. Thus $A_m \leq 2\sqrt{m}$. Combining this deterministic bound with (8), and combining the uniform second-moment bound for W_m with Cauchy–Schwarz, gives

$$\mathbb{E}[A_m \mathbf{1}_{G_m^c}] \longrightarrow 0, \quad \mathbb{E}[W_m \mathbf{1}_{G_m^c}] \longrightarrow 0.$$

The sandwich (9), $\delta_m \rightarrow 0$, and (7) therefore imply

$$\mathbb{E}A_m \longrightarrow \sqrt{\frac{\pi}{2}}. \quad (10)$$

Using (5) and $m = n + 1$, we obtain

$$2 - \mathbb{E}\|X + Y\|_K = m\mathbb{E}M_m = \frac{\mathbb{E}A_m}{\sqrt{m}} = \sqrt{\frac{\pi}{2n}} + o(n^{-1/2}),$$

which proves (3). Comparing with (1) proves the last assertions. $\square$

4 Verification and scope

The accompanying script numerically integrates the exact survival function $q(t/\sqrt{m})^m$ and independently samples the normalized Dirichlet model. Those computations check the scaling and constants but are not used in the proof.

The proof establishes the desired order for centered polytopes with $O(n)$ facets and the exact leading term for simplices. It is consistent with, but does not prove, a universal deficit of order $n^{-1/2}$ for bodies with arbitrary or infinitely many supporting facets. It also shows that the cube is not a maximizer of $\mathbb{E}\|X + Y\|_K$ in high dimension, should one consider that stronger variant. The published source passage itself asks for the order of the worst deficit and does not require cube extremality.

A bounded novelty check searched the local run indexes, the parsed arXiv corpus, arXiv, and exact web phrases involving uniform simplex points, Minkowski gauges of $X + Y$, and minima of sums of flat Dirichlet coordinates. No prior exact computation was found. This is not an exhaustive literature claim.

References

- [1] H. Huang, B. A. Slomka, T. Tkocz, and B.-H. Vritsiou, *Improved bounds for Hadwiger’s covering problem via thin-shell estimates*, arXiv:1811.12548, especially Section 6 (page 15).